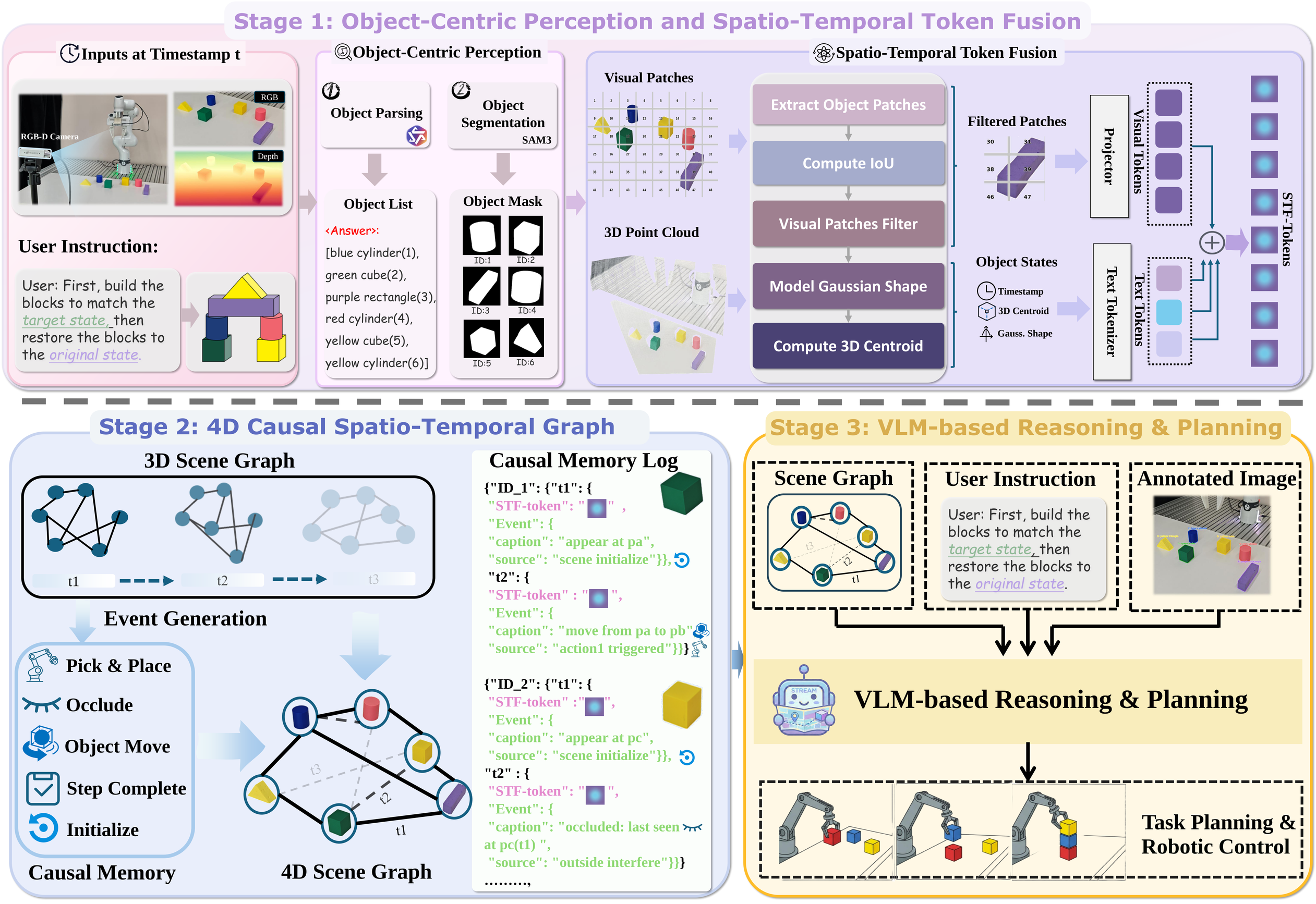
Stage 1: Object-Centric Perception and Spatio-Temporal Token Fusion
Inputs at Timestamp t
RGB-D Camera
RGB
Depth
User Instruction:
User: First, build the blocks to match the target state, then restore the blocks to the original state.
Object-Centric Perception
Object Parsing
Object Segmentation
SAM3
Object List
<Answer>:
[blue cylinder(1), green cube(2), purple rectangle(3), red cylinder(4), yellow cube(5), yellow cylinder(6)]
Object Mask
ID:1
ID:2
ID:3
ID:4
ID:5
ID:6
Spatio-Temporal Token Fusion
Visual Patches
3D Point Cloud
Extract Object Patches
Compute IoU
Visual Patches Filter
Model Gaussian Shape
Compute 3D Centroid
Filtered Patches
Object States
Timestamp
3D Centroid
Gauss. Shape
Projector
Text Tokenizer
Visual Tokens
Text Tokens
STF-Tokens
Stage 2: 4D Causal Spatio-Temporal Graph
3D Scene Graph
t1
t2
t3
Event Generation
Pick & Place
Occlude
Object Move
Step Complete
Initialize
Causal Memory
4D Scene Graph
Causal Memory Log
{"ID_1": {"t1": {
"STF-token": "
"Event": {
"caption": "appear at pa",
"source": "scene initialize"}},
"t2": {
"STF-token" : "
"Event": {
"caption": "move from pa to pb"
"source": "action1 triggered"}}}
{"ID_2": {"t1": {
"STF-token" :"
"Event": {
"caption": "appear at pc",
"source": "scene initialize"}},
"t2" : {
"STF-token": "
"Event": {
"caption": "occluded: last seen at pc(t1) ",
"source": "outside interfere"}}}
.....,
Stage 3: VLM-based Reasoning & Planning
Scene Graph
User Instruction
User: First, build the blocks to match the target state, then restore the blocks to the original state.
Annotated Image
VLM-based Reasoning & Planning
Task Planning & Robotic Control